
SRI VENKATESWARA COLLEGE OF ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Subject: Data Structures and Algorithms

Subject Code: CS3301

Semester: III

Regulation: R2021

Exam Type: Internal Assessment - I

Max. Marks: 100

Duration: 3 Hours

Date: 12-04-2026

PART – A (2 Marks)

Answer ALL Questions (10 × 2 = 20 Marks)

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| 1. Define data structure and classify its types. | [2] | K1 |
| 2. State the difference between linear and non-linear data structures. | [2] | K1 |
| 3. What is a stack? Mention its basic operations. | [2] | K1 |
| 4. List the applications of queue in real-time systems. | [2] | K1 |
| 5. Define a linked list. State its advantages over arrays. | [2] | K1 |
| 6. Write the postfix notation for the expression (A+B)*(C-D). | [2] | K2 |
| 7. State the properties of a binary tree. | [2] | K1 |
| 8. Define graph and mention its types. | [2] | K1 |
| 9. What is hashing? List any two collision resolution techniques. | [2] | K1 |
| 10. Mention the difference between BFS and DFS traversal. | [2] | K2 |
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PART – B (16 Marks)

Answer any FIVE Questions (5 × 16 = 80 Marks)

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| 11. Explain the array representation of stack and queue with suitable examples. Discuss push, pop, enqueue, and dequeue operations with algorithms. | [16] | K2 |
| 12. Describe the singly linked list. Write algorithms for the following operations: | [16] | K2 |
| (a) Insertion at the beginning, middle, and end of a linked list. | [8] | K2 |
| (b) Deletion of a node from any given position. | [8] | K2 |
| 13. Analyze the time and space complexity of the following sorting algorithms: | [16] | K4 |
| (a) Bubble Sort | [4] | K4 |
| (b) Selection Sort | [4] | K4 |
| (c) Insertion Sort | [4] | K4 |
| (d) Merge Sort | [4] | K4 |
| 14. Explain binary search tree (BST). Construct a BST for the elements: 50, 30, 70, 20, 40, 60, 80. Perform in-order, pre-order, and post-order traversal on the constructed tree. | [16] | K2 |
| 15. Design a hash table of size 10 using chaining method for the keys: 25, 42, 96, 101, 102, 162, 197. Analyze performance and compare with open addressing. | [16] | K6 |
| 16. Compare BFS and DFS graph traversal. Apply both on graph $V=\{A,B,C,D,E\}$, $E=\{AB,AC,BD,BE,CD,CE,DE\}$. Draw the traversal order. | [16] | K4 |

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| 17. | Discuss AVL trees and their rotations with examples. | [16] | K2 |
| (a) | Explain LL, RR, LR, and RL rotations with diagrams. | [8] | K2 |
| (b) | Construct an AVL tree inserting: 10, 20, 30, 40, 50, 25. | [8] | K3 |
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*** END OF QUESTION PAPER ***